
Software Requirements Specification

for

Inventory Management System for Heshan Foods

Version 1.0 approved

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18/02/2026

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Revision History

Name	Date	Reason For Changes	Version
Uditha Balalla	05/02/2026	Initial draft of Software Requirements Specification	0.1
Uditha Balalla	10/02/2026	Added functional requirements and system features	0.1
Uditha Balalla	15/02/2026	Added non-functional requirements and appendices	0.9
Uditha Balalla	18/02/2026	Final review, formatting, and approval for submission	1.0 (Approved)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the requirements for version 1.0 (Approved) of the Factory Inventory & Production Management System, developed as part of the Agile Project Management module at the University of Bedfordshire by Lumora Software Solutions..

The purpose of this document is to define the functional and non-functional requirements of the system, which supports the management of inventory, production processes, order handling, and dispatch operations within a single factory environment. The system has been implemented as a Minimum Viable Product (MVP) and represents the fully developed version delivered during the project lifecycle.

- This SRS covers the complete software solution, including:
- The web-based client application
- The backend server and API services
- The database layer
- Role-based access control mechanisms
- Reporting and audit functionalities

The document serves as a formal reference for stakeholders including developers, testers, academic assessors, and project team members. It establishes a clear understanding of system behaviour, constraints, and quality expectations in alignment with Agile project management principles.

1.2 Document Conventions

This Software Requirements Specification follows the IEEE-recommended structure for SRS documentation.

The following conventions are used throughout this document:

- All mandatory requirements are expressed using the keyword **"SHALL"** to indicate compulsory system behaviour.
- Functional requirements are uniquely identified using structured requirement identifiers in the format:
 - **F-INV-XX** → Inventory-related functional requirements
 - **F-PROD-XX** → Production-related functional requirements
 - **F-ORD-XX** → Order-related functional requirements
 - **F-DIS-XX** → Dispatch-related functional requirements
 - **F-REP-XX** → Reporting requirements
 - **NF-XX** → Non-functional requirements

- Each requirement is uniquely numbered to ensure traceability and ease of validation.
- Requirements are written to be:
 - Clear
 - Testable
 - Unambiguous
 - Necessary
 - Verifiable

Where information is pending clarification, the term **TBD (To Be Determined)** will be used.

Higher-level requirements do not automatically imply priority inheritance; each requirement is independently stated and defined.

1.3 Intended Audience and Reading Suggestions

This Software Requirements Specification (SRS) is intended for the following stakeholders:

- **Developers** – responsible for implementing, modifying, and maintaining the system.
- **Testers** – responsible for verifying that the implemented system satisfies all defined requirements.
- **Project Team Members** – involved in planning, coordination, documentation, and evaluation of the project deliverables.

This document provides a structured and comprehensive definition of the system requirements. It is organized as follows:

- Section 1 introduces the purpose, scope, and conventions used within this document.
- Section 2 provides an overall description of the system context, user classes, and operational constraints.
- Section 3 defines external interface requirements.
- Section 4 details the system features and associated functional requirements.
- Section 5 specifies non-functional requirements including performance, security, and quality attributes.
- Appendices provide supporting material such as glossary terms and analysis models.

Readers are encouraged to review the Introduction and Overall Description sections before proceeding to detailed functional requirements. Developers and testers should particularly focus on Sections 3, 4, and 5, which define system behaviour and validation criteria.

1.4 Product Scope

The Factory Inventory & Production Management System is a web-based software solution developed to support and optimise operational processes within a single factory environment.

The primary objective of the system is to digitise and streamline inventory and production workflows while improving traceability, stock control accuracy, and operational efficiency. The system replaces manual or fragmented processes with a structured, role-based digital platform that manages the full lifecycle of:

- Raw material intake (Goods Received Notes – GRN)
- Production batch management
- Finished goods inventory tracking
- Order processing
- Dispatch allocation using FEFO (First-Expiry-First-Out)
- Reporting and audit tracking

The system enables clearly defined user roles (Owner, Employee, Wholesaler) to interact with operational data securely and efficiently through role-based access control mechanisms.

By integrating inventory, production, and dispatch workflows into a single system, the product supports:

- Improved stock visibility and traceability
- Reduction of inventory discrepancies
- Controlled production lifecycle management
- Enhanced accountability through audit logging
- Data-driven reporting for operational insight

The delivered version represents a fully implemented Minimum Viable Product (MVP), developed using Agile project management principles and iterative delivery.

2. Overall Description

2.1 Product Perspective

The Factory Inventory & Production Management System is a newly developed, standalone web-based application designed to replace previously manual and fragmented operational processes within a single factory environment.

Prior to system implementation, inventory tracking, production lifecycle management, and dispatch operations were assumed to be handled through paper-based records and disconnected spreadsheet tools. These approaches introduced limitations including:

- Limited traceability
- Increased risk of human error
- Delayed reporting
- Poor visibility of stock levels
- Inconsistent audit accountability

The developed system centralises these operational processes into an integrated digital platform that provides structured workflow control, real-time data visibility, and enforced role-based access control.

The system architecture follows a client-server model consisting of:

- A web-based frontend application
- A backend API service
- A centralised database
- Authentication services

The product does not act as a subsystem of a larger enterprise platform; instead, it operates as an independent operational management system tailored to a single factory context.

The system interfaces with external services for authentication but maintains full control over operational, production, and inventory data internally.

2.2 Product Functions

The Factory Inventory & Production Management System provides the following high-level functional capabilities:

1. Inventory Management

- Record and confirm Goods Received Notes (GRN) for raw materials.
- Maintain raw material stock levels.
- Track finished goods inventory by batch.
- Record stock movements for traceability.
- Enforce FEFO (First-Expiry-First-Out) allocation logic during dispatch.

2. Production Management

- Create and manage Bills of Materials (BOM).
- Create, start, and complete production batches.
- Record raw material consumption during production.
- Record finished goods output and wastage.
- Track production lifecycle states.

3. Order and Dispatch Management

- Create and manage customer (wholesaler) orders.
- Confirm and cancel orders.
- Allocate finished goods to confirmed orders.
- Generate dispatch records.
- Produce delivery note documentation in PDF format.

4. Reporting and Traceability

- Generate stock level reports.
- Provide stock movement history.
- Provide expiry tracking visibility.
- Generate production and wastage reports.
- Enable traceability from raw material intake to finished goods dispatch.
- Maintain audit logs of system actions.

5. User and Access Control

- Authenticate users using secure credentials.
- Assign role-based permissions (Owner, Employee, Wholesaler).
- Restrict access to system modules based on user roles.
- Log user actions for accountability.

2.3 User Classes and Characteristics

The system supports three primary user classes, each with distinct responsibilities and access privileges.

1. Owner

The Owner is the highest-privileged user within the system.

- **Technical proficiency:** Medium
- **Frequency of use:** Daily
- **Authority level:** Full system control

The Owner has unrestricted access to all modules including user administration, inventory management, production management, order processing, reporting, and audit logs. This user class is responsible for overall system oversight, configuration of master data, and operational supervision.

The Owner is considered a critical user class due to their governance and decision-making responsibilities.

2. Employee

Employees are operational users responsible for executing daily production and inventory tasks.

- **Technical proficiency:** Medium
- **Frequency of use:** Daily
- **Operational focus:** Production and inventory management

Employees interact with modules such as GRN confirmation, batch lifecycle management, BOM maintenance, order confirmation, and dispatch creation. They do not have authority to manage user accounts or access audit logs.

This user class is essential for maintaining operational continuity and system data accuracy.

3. Wholesaler

Wholesalers are external-facing users who interact with the system for order-related activities only.

- **Technical proficiency:** Medium
- **Frequency of use:** Occasional
- **Functional scope:** Order creation and order status viewing

Wholesalers are restricted to creating orders, viewing the status of their own orders, and accessing delivery note documentation related to their dispatches. They do not have access to inventory, production, or internal reporting modules.

This user class has limited privileges and interacts with a restricted subset of system functionality.

2.4 Operating Environment

The Factory Inventory & Production Management System operates within a web-based client–server architecture.

Client Environment

- Web-based frontend developed using **React (Vite build system)**.
- Accessible through modern web browsers (e.g., Google Chrome, Microsoft Edge, Firefox).
- No local installation required.
- Internet connection required for system access.
- Designed for desktop and laptop environments.

Server Environment

- Backend developed using **Node.js with Express framework**.
- Deployed as a serverless API environment.
- Hosted using a cloud-based deployment platform.
- Implements RESTful API architecture.

Database Environment

- Centralised database hosted on **MongoDB Atlas**.
- Replica-set configuration required to support transaction-based operations.
- Stores master data, operational records, stock movements, and audit logs.

Authentication Environment

- User authentication handled via **Firebase Authentication (Email/Password model)**.
- Backend verifies Firebase-issued tokens using Firebase Admin SDK.
- Role-based access control enforced within the backend application.

Deployment Environment

- Frontend and backend deployed as separate projects using cloud hosting services.
- Communication between client and server occurs over HTTPS.
- Cross-Origin Resource Sharing (CORS) configuration enforced for secure access.

2.5 Design and Implementation Constraints

The development and implementation of the Factory Inventory & Production Management System were subject to the following constraints:

1. Technology Constraints

The system was required to be developed using specific technologies selected by the project team:

- Frontend: React (Vite)
- Backend: Node.js with Express
- Database: MongoDB Atlas
- Authentication: Firebase Authentication
- Deployment: Cloud-based hosting platform

These technology choices constrained architectural decisions, data modelling approaches, and integration patterns.

2. Methodological Constraint

The system was developed within the framework of an Agile project management methodology. Iterative development cycles, sprint reviews, and incremental feature delivery influenced prioritisation and scope decisions throughout the project lifecycle.

3. Time Constraint

The project was completed within a fixed academic timeframe of approximately ten weeks. This limited the scope of advanced feature implementation and necessitated prioritisation of core functionality consistent with a Minimum Viable Product (MVP) delivery model.

4. Academic Environment Constraint

As the system was developed within a university academic setting, access to enterprise infrastructure, commercial tooling, and extended production testing environments was limited.

5. Security and Access Control Constraint

The system was required to implement Role-Based Access Control (RBAC) to enforce user privilege separation between Owner, Employee, and Wholesaler roles. Security considerations influenced API design and module access restrictions.

6. Database Transaction Constraint

Operational processes such as stock movement, batch completion, and dispatch allocation required transactional consistency. Therefore, the database environment required support for replica-set configuration to enable atomic transaction handling.

7. Deployment Dependency Constraint

The system depends on cloud-based services for hosting, authentication, and database management. Reliable internet connectivity and correct environment configuration are required for proper operation.

8. Platform Scope Constraint

The delivered system is a browser-based web application only. No dedicated mobile or desktop-native application was developed within the scope of this project.

2.6 User Documentation

The following documentation is provided to support system usage and deployment:

1. User Guide

A structured User Guide is provided to describe:

- System login procedures
- Role-based access capabilities (Owner, Employee, Wholesaler)
- Workflow instructions for:
 - Creating and confirming GRNs
 - Managing production batches
 - Processing and confirming orders
 - Creating dispatches
 - Generating delivery note PDFs
 - Accessing reports and audit logs

The User Guide is intended for operational users and provides step-by-step instructions for interacting with the system.

2. Technical Setup Documentation

Technical documentation is provided for:

- Local development environment setup
- Environment variable configuration
- Database connection requirements
- Authentication configuration
- Cloud deployment steps

This documentation is intended for developers and maintainers.

3. Deployment Guide

A deployment guide describes:

- Cloud hosting configuration
- Environment variable requirements
- CORS configuration
- Pre-deployment validation steps

This ensures that the system can be redeployed or maintained in future environments.

4. In-System Interface Guidance

The system includes:

- Structured navigation
- Clear module labelling
- Role-restricted menu visibility
- Form validation messages
- Error feedback responses

These interface features reduce reliance on external documentation.

2.7 Assumptions and Dependencies

The following assumptions and external dependencies apply to the Factory Inventory & Production Management System:

Assumptions

1. **User Digital Literacy**

It is assumed that all users (Owner, Employee, Wholesaler) possess at least a medium level of digital proficiency and are capable of interacting with a web-based system.

2. **Stable Internet Connectivity**

The system assumes reliable internet access for all users, as it operates entirely as a cloud-based web application.

3. **Accurate Data Entry**

It is assumed that users will enter stock quantities, batch details, and order data accurately. The system enforces validation rules but relies on correct human input for operational integrity.

4. **Single Factory Operation**

The system is designed for a single factory context. Multi-site operations, distributed warehouses, or cross-location stock transfers are outside the scope of this MVP.

5. **Standalone Operational Use**

The system operates independently and is not assumed to integrate with external ERP, accounting, or supply chain systems within the current scope.

6. **MVP Scope Limitation**

The delivered system represents a Minimum Viable Product. Advanced analytics, automation features, or enterprise scalability enhancements are assumed to be future improvements.

Dependencies

1. **Firebase Authentication Service**

User authentication depends on the availability and correct configuration of Firebase Authentication services.

2. **MongoDB Atlas Database Service**

System data storage and transaction handling depend on MongoDB Atlas, including replica-set configuration for transaction support.

3. **Cloud Hosting Infrastructure**

Deployment and system availability depend on cloud hosting services for both frontend and backend components.

4. Browser Compatibility

The system depends on modern browser environments that support JavaScript modules and HTTPS communication.

3. External Interface Requirements

3.1 User Interfaces

The system provides a web-based graphical user interface (GUI) accessible via modern web browsers. No desktop installation is required.

3.1.1 General Interface Characteristics

- The system follows a consistent layout structure consisting of:
 - A left-hand navigation panel (collapsible)
 - A top header bar
 - A central content display area
- All pages use consistent typography, spacing, and visual hierarchy.
- Interface components include:
 - Data tables
 - Modal dialog forms
 - Dropdown selectors
 - Date pickers
 - Numeric input controls
 - Status tags
 - Confirmation buttons

The interface supports both light and dark display modes.

3.1.2 Authentication Interface

The system provides:

- A login screen requiring:
 - Email address (validated format)
 - Password
- Validation for required fields
- Error feedback for invalid credentials
- A restricted access screen for authenticated users not registered in the system database

Upon successful authentication, users are redirected to the system dashboard.

3.1.3 Navigation Structure

The interface includes a role-based navigation menu.

Menu visibility is dynamically controlled according to user role:

- Owner: Full system access including Users and Audit modules
- Employee: Operational modules (Inventory, Production, Orders, Dispatch, Reports)
- Wholesaler: Limited to Orders and dispatch documentation

Unauthorized access attempts result in redirection to an allowed route.

3.1.4 Core Interface Screens

The system provides the following primary interface screens:

1. Dashboard – Displays summary metrics and system statistics.
 2. Products – Allows creation and management of raw materials and finished goods.
 3. Suppliers – Allows supplier data management.
 4. Wholesalers – Allows wholesaler data management.
 5. GRN – Supports creation and confirmation of goods received notes.
 6. BOM – Allows definition of bill of materials for finished goods.
 7. Batches – Manages production lifecycle (Draft → In Progress → Completed).
 8. Orders – Supports order creation, confirmation, and cancellation.
 9. Dispatches – Enables dispatch creation and delivery note generation.
 10. Reports – Provides tab-based operational and analytical reports.
 11. Users – Owner-only user management interface.
 12. Audit Log – Owner-only system audit interface.
-

3.1.5 Form and Validation Standards

- All required fields are clearly marked.
 - Numeric fields enforce minimum value constraints.
 - Email fields validate proper format.
 - Date fields enforce logical sequencing (e.g., expiry date after manufacture date).
 - Forms prevent submission when validation fails.
 - Success and error feedback is displayed via notification messages.
-

3.1.6 Status Representation

System entities use visual status indicators including:

- DRAFT
- CONFIRMED
- IN_PROGRESS
- COMPLETED
- DISPATCHED
- CANCELLED

Status transitions are controlled through action buttons and are validated by the backend before confirmation.

3.1.7 Error Handling and Feedback

- Loading indicators are displayed during API calls.
- Error messages are presented in a consistent notification format.
- Unhandled application errors are captured and displayed via a fallback error screen.
- Authentication failure results in redirection to the login page.

3.2 Hardware Interfaces

The Factory Inventory & Production Management System does not directly interface with any specialised hardware devices. The system operates entirely as a web-based application and requires only standard computing hardware capable of running a modern web browser.

Supported Hardware Environment

- Desktop or laptop computer
- Standard keyboard and mouse input devices
- Internet connectivity
- Standard display monitor

The system does not require or integrate with:

- Barcode scanners
- RFID readers
- IoT production devices
- Industrial machinery controllers
- Dedicated printing hardware
- Local server infrastructure

All processing is handled through cloud-hosted services, and interaction occurs through standard HTTP/HTTPS communication between client and server.

3.3 Software Interfaces

The Factory Inventory & Production Management System interfaces with the following external software components:

1. Firebase Authentication

The system integrates with Firebase Authentication for user identity verification.

- Authentication model: Email and Password
- The client application obtains an authentication token upon successful login.
- The backend verifies the token using Firebase Admin SDK.
- User role information is retrieved from the internal system database after authentication.

Data exchanged:

- Authentication tokens (JWT)
- User email and unique Firebase UID

The system depends on correct configuration of Firebase project credentials.

2. MongoDB Atlas Database

The system interfaces with MongoDB Atlas for persistent data storage.

The database stores:

- User records
- Products
- Suppliers
- Wholesalers
- GRN records
- BOM definitions
- Batch records
- Orders
- Dispatch records
- Stock movement logs
- Audit logs

Transactions are used for operations requiring atomicity (e.g., batch completion and dispatch allocation). A replica-set configuration is required to support transactional consistency.

3. Cloud Hosting Platform

The application is deployed using a cloud-based hosting environment.

- The frontend and backend are deployed as separate services.
 - Communication occurs via HTTPS RESTful API calls.
 - Cross-Origin Resource Sharing (CORS) configuration restricts access to approved origins.
-

4. Web Browser Environment

The system interfaces with standard web browsers that support:

- JavaScript ES modules
- HTTPS communication
- Local storage for theme preference persistence

The browser environment executes the client application and renders the graphical user interface.

5. PDF Generation Library

The backend integrates with a PDF generation library to produce delivery note documents for dispatch records.

Generated PDFs include:

- Dispatch details
- Order information
- Product allocations
- Batch references

The PDF is returned as a downloadable file via HTTP response.

The system does not integrate with:

- Enterprise Resource Planning (ERP) systems
- Accounting systems
- External supply chain platforms
- Payment gateways

All operational logic is internally managed.

3.4 Communications Interfaces

The Factory Inventory & Production Management System communicates using standard web-based protocols over secure internet connections.

3.4.1 Communication Protocol

- All client-server communication occurs over **HTTPS**.
- The system follows a **RESTful API architecture**.
- No SOAP-based services are used.
- No WebSocket communication is implemented.

All API endpoints are accessed via secure HTTP methods, including:

- GET
 - POST
 - PUT
 - PATCH
 - DELETE
 - OPTIONS (for preflight handling)
-

3.4.2 Data Format

- All data exchanged between the client and server is formatted using **JSON (JavaScript Object Notation)**.
 - Authentication tokens are transmitted in the HTTP **Authorization** header using the Bearer token scheme.
 - PDF delivery notes are transmitted as binary file responses over HTTPS.
-

3.4.3 Security Mechanisms

- All communications are encrypted using HTTPS (TLS).
 - Cross-Origin Resource Sharing (CORS) policies restrict access to approved client origins.
 - Authentication tokens are validated server-side before processing requests.
 - Role-based access control is enforced after successful authentication.
-

3.4.4 Communication Scope Limitations

The system does not support:

- Email-based communication
- FTP file transfers
- Messaging queues
- Real-time socket-based communication
- External API integrations

All operational communication occurs strictly between the web client and the backend server via RESTful HTTP calls.

4. System Features

4.1 Inventory Management

4.1.1 Description and Priority

The Inventory Management feature enables management of products, suppliers, wholesalers, and Goods Received Notes (GRN). It maintains raw material stock levels and records inventory movements.

Priority: High

Inventory accuracy is essential for production and dispatch operations.

4.1.2 Stimulus/Response Sequences

- **Stimulus 1:** User creates a product.
Response: System validates required fields and stores the product record.
- **Stimulus 2:** User creates a GRN.
Response: System stores GRN in DRAFT status.
- **Stimulus 3:** User confirms a GRN.
Response: System validates status, updates stock quantities, records stock movement, and sets GRN status to CONFIRMED.
- **Stimulus 4:** User attempts to modify a confirmed GRN.
Response: System rejects modification and displays an error message.

4.1.3 Functional Requirements

- **F-INV-01**
The system SHALL allow authorised users to create and manage product records.
- **F-INV-02**
The system SHALL allow authorised users to create and manage supplier records.
- **F-INV-03**
The system SHALL allow authorised users to create and manage wholesaler records.
- **F-INV-04**
The system SHALL create GRNs in DRAFT status.
- **F-INV-05**
The system SHALL allow confirmation of GRNs only when status is DRAFT.
- **F-INV-06**
Upon GRN confirmation, the system SHALL update raw material stock quantities.
- **F-INV-07**
The system SHALL record stock movement entries for inventory changes.

- **F-INV-08**
The system SHALL prevent modification of confirmed GRNs.

4.2 Production Management

4.2.1 Description and Priority

The Production Management feature enables definition of Bills of Materials (BOM) and management of production batch lifecycles, including raw material consumption, finished goods creation, and wastage recording.

Priority: High

Production operations directly impact inventory accuracy and order fulfilment.

4.2.2 Stimulus/Response Sequences

- **Stimulus 1:** User creates a BOM.
Response: System validates inputs and stores component relationships.
 - **Stimulus 2:** User creates a production batch.
Response: System stores batch in DRAFT status.
 - **Stimulus 3:** User starts a batch.
Response: System updates batch status to IN_PROGRESS.
 - **Stimulus 4:** User completes a batch.
Response: System validates stock availability, deducts raw materials, creates finished goods stock, records wastage, updates batch status to COMPLETED, and logs stock movements.
-

4.2.3 Functional Requirements

- **F-PROD-01**
The system SHALL allow authorised users to create BOM records.
- **F-PROD-02**
The system SHALL allow creation of production batches in DRAFT status.
- **F-PROD-03**
The system SHALL allow batch status transitions (DRAFT → IN_PROGRESS → COMPLETED).
- **F-PROD-04**
Upon batch completion, the system SHALL deduct raw material stock.
- **F-PROD-05**
Upon batch completion, the system SHALL create finished goods stock records.
- **F-PROD-06**
The system SHALL allow recording of production wastage.

- **F-PROD-07**

The system SHALL execute batch completion operations transactionally.

4.3 Order Management

4.3.1 Description and Priority

The Order Management feature enables creation, confirmation, and cancellation of wholesaler orders with controlled lifecycle transitions.

Priority: High

Orders drive dispatch and revenue generation.

4.3.2 Stimulus/Response Sequences

- **Stimulus 1:** User creates an order.
Response: System stores order in DRAFT status.
 - **Stimulus 2:** Authorised user confirms an order.
Response: System updates order status to CONFIRMED.
 - **Stimulus 3:** Authorised user cancels an order.
Response: System updates order status to CANCELLED.
-

4.3.3 Functional Requirements

- **F-ORD-01**
The system SHALL allow authorised users to create orders containing finished goods.
- **F-ORD-02**
The system SHALL assign newly created orders a DRAFT status.
- **F-ORD-03**
The system SHALL allow authorised users to confirm DRAFT orders.
- **F-ORD-04**
The system SHALL allow authorised users to cancel orders.
- **F-ORD-05**
The system SHALL prevent dispatch for orders not in CONFIRMED status.

4.4 Dispatch Management

4.4.1 Description and Priority

The Dispatch Management feature enables allocation of finished goods to confirmed orders using FEFO (First-Expiry-First-Out) logic and generates delivery note documentation.

Priority: High

Dispatch ensures accurate stock reduction and order fulfilment.

4.4.2 Stimulus/Response Sequences

- **Stimulus 1:** User creates a dispatch for a confirmed order.
Response: System validates order status, allocates stock using FEFO, deducts finished goods stock, records stock movement, and creates dispatch records.
 - **Stimulus 2:** User downloads delivery note.
Response: System generates and returns a PDF document.
-

4.4.3 Functional Requirements

- **F-DIS-01**
The system SHALL allow dispatch creation only for CONFIRMED orders.
- **F-DIS-02**
The system SHALL allocate finished goods using FEFO logic.
- **F-DIS-03**
Upon dispatch creation, the system SHALL deduct finished goods stock.
- **F-DIS-04**
The system SHALL record stock movement entries for dispatch operations.
- **F-DIS-05**
The system SHALL generate a delivery note PDF for each dispatch.
- **F-DIS-06**
The system SHALL prevent duplicate dispatch creation for the same order.

4.5 Reporting and Traceability

4.5.1 Description and Priority

The Reporting and Traceability feature provides visibility into stock levels, movements, expiry, production, wastage, sales, and audit activities.

Priority: Medium

Reporting supports monitoring and operational decision-making.

4.5.2 Stimulus/Response Sequences

- **Stimulus 1:** User selects a report and applies filters.
Response: System retrieves filtered data and displays results.
 - **Stimulus 2:** User performs traceability search.
Response: System displays linked records from intake to dispatch.
-

4.5.3 Functional Requirements

- **F-REP-01**
The system SHALL provide stock-on-hand reporting.
- **F-REP-02**
The system SHALL provide stock movement history.
- **F-REP-03**
The system SHALL provide expiry tracking visibility.
- **F-REP-04**
The system SHALL provide production and wastage reports.
- **F-REP-05**
The system SHALL provide sales-by-wholesaler reporting.
- **F-REP-06**
The system SHALL provide traceability from raw material intake to dispatch.

4.6 User and Access Control

4.6.1 Description and Priority

The User and Access Control feature manages authentication and enforces role-based permissions across system modules.

Priority: High

Access control protects operational data and system integrity.

4.6.2 Stimulus/Response Sequences

- **Stimulus 1:** User logs in with valid credentials.
Response: System authenticates user and loads role-based permissions.
 - **Stimulus 2:** User attempts to access restricted module.
Response: System denies access and redirects appropriately.
-

4.6.3 Functional Requirements

- **F-SEC-01**
The system SHALL authenticate users using secure credentials.
- **F-SEC-02**
The system SHALL assign role-based permissions (Owner, Employee, Wholesaler).
- **F-SEC-03**
The system SHALL restrict module access based on user role.
- **F-SEC-04**
The system SHALL log user actions in an audit log.
- **F-SEC-05**
The system SHALL terminate user session upon logout.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- **NF-01**
The system SHALL respond to standard user actions within 3 seconds under normal operating conditions.
- **NF-02**
The system SHALL support concurrent access by multiple authenticated users without data corruption.
- **NF-03**
Inventory, batch completion, and dispatch operations SHALL execute atomically to prevent inconsistent stock states.
- **NF-04**
Report queries SHALL return results within acceptable response time for typical dataset sizes in a single-factory environment.

5.2 Safety Requirements

The system does not control physical machinery and therefore does not introduce direct physical safety risks. However:

- **NF-05**
The system SHALL prevent negative stock quantities.
- **NF-06**
The system SHALL prevent invalid production or dispatch operations that could lead to operational inconsistencies.

5.3 Security Requirements

- **NF-07**
The system SHALL require authentication before granting access to protected modules.
- **NF-08**
The system SHALL enforce role-based access control (RBAC).
- **NF-09**
The system SHALL transmit all data over HTTPS.
- **NF-10**
The system SHALL validate authentication tokens server-side before processing requests.
- **NF-11**
The system SHALL record audit logs for critical operations.

5.4 Software Quality Attributes

- **NF-12 (Reliability)**
The system SHALL maintain data integrity during transactional operations.

- **NF-13 (Usability)**
The system SHALL provide consistent interface layout and validation feedback.
- **NF-14 (Maintainability)**
The system SHALL follow modular architecture separating frontend, backend, and database layers.
- **NF-15 (Availability)**
The system SHALL depend on cloud infrastructure availability.
- **NF-16 (Scalability – Limited Scope)**
The system SHALL support scaling within a single-factory operational context.

5.5 Business Rules

- **BR-01**
Only DRAFT GRNs may be confirmed.
- **BR-02**
Only CONFIRMED orders may be dispatched.
- **BR-03**
Batch status transitions SHALL follow the sequence:
DRAFT → IN_PROGRESS → COMPLETED.
- **BR-04**
FEFO (First-Expiry-First-Out) SHALL be applied during dispatch allocation.
- **BR-05**
Only Owners may manage user accounts and access audit logs.

6. Other Requirements

The system SHALL maintain structured collections for:

- Users
- Products
- Suppliers
- Wholesalers
- GRNs
- BOMs
- Batches
- Orders
- Dispatches
- Stock Movements
- Audit Logs

The database SHALL support transactional operations to ensure data consistency during batch completion and dispatch processing.

6.2 Legal and Ethical Considerations

- The system SHALL protect user authentication data through secure credential handling.
 - The system SHALL restrict access to operational data based on role-based permissions.
 - The system SHALL log critical actions to ensure accountability and traceability.
 - No personal financial or payment data is processed by the system.
-

6.3 Future Enhancements (Out of Scope)

The following features are outside the scope of the current MVP:

- Multi-factory support
- Integration with ERP or accounting systems
- Mobile-native application
- Automated hardware integration

Appendix A: Glossary

Term	Definition
GRN	<i>Goods Received Note – record of raw material intake</i>
BOM	<i>Bill of Materials – list of raw materials required per finished product</i>
Batch	<i>Production unit representing a manufacturing cycle</i>
FEFO	<i>First-Expiry-First-Out – stock allocation method prioritising earliest expiry</i>
RBAC	<i>Role-Based Access Control</i>
DRAFT	<i>Initial state before confirmation</i>
CONFIRMED	<i>Approved operational state</i>
IN_PROGRESS	<i>Active production state</i>
COMPLETED	<i>Finalised production state</i>
DISPATCHED	<i>Order fulfilled and stock allocated</i>

Appendix B: Analysis Models

The system architecture follows a client–server model consisting of:

- Web-based client interface
- RESTful backend API
- Centralised cloud-hosted database
- External authentication provider

Core lifecycle models include:

- GRN: DRAFT → CONFIRMED
- Batch: DRAFT → IN_PROGRESS → COMPLETED
- Order: DRAFT → CONFIRMED → DISPATCHED / CANCELLED

Appendix C: To Be Determined List

No outstanding TBD items remain in the current version (1.0 Approved).

References

6.1 References

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